

Po-Chun Wu

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Education

- **Carnegie Mellon University** Pittsburgh, PA
M.S. Electrical & Computer Engineering / ETIM Dual Degree Expected May 2028
- **National Taiwan University** Taipei, Taiwan
B.S. Electrical Engineering; GPA: 4.16/4.3 Aug 2022 – Jan 2026
 - **Courses:** Data Structures, Algorithms, Electronic Design Automation, SoC Verification, Machine Learning

Experience

- **Software Engineer, SLEKMED — Taipei, Taiwan** Aug 2023 – Mar 2026
 - **Note-to-Video Generator:** Developed a web-based note-to-video generator pipeline using Marp, OpenAI API, and FFmpeg, enabling medical experts to create educational videos from their notes.
 - **Medical Simulation Platform:** Built Next.js simulation modules for medical diagnostic training in collaboration with Taipei Medical University peers, gaining 4,200+ followers and securing funding from the Taiwan Ministry of Education.
 - **Virtual Patient Chatbot:** Implemented an LLM-based chatbot as a virtual patient, contributing to a 1st-place finish among 40+ teams in the 2024 CTCI AI Competition (12K USD prize).
- **Co-founder, Synality — Taipei, Taiwan** Dec 2024 – Feb 2025
 - **Startup Development:** Built a full-stack Next.js matchmaking app within 10 days, conducted three rounds of minimum viable product testing with 60 participants, and improved contact conversion from 24% to 45%.

Projects

- **Whiskers IDE** Jan 2026 – Feb 2026
 - Built a Next.js-based compiler to convert a Scratch-inspired programming syntax into JavaScript.
 - Designed a drag-and-drop interface to facilitate new learners' transition from Scratch to text-based programming.
- **Formal Verification (Advising: Prof. Chung-Yang Ric Huang)** Apr 2025 – Jun 2025
 - Modeled optimal vocabulary learning sequence generation as an NP-hard Target Set Selection variant.
 - Implemented bounded model checking-based verification to derive optimal solutions for graphs with 150 nodes.
- **LifeOS Automation System** Jan 2025 – Mar 2025
 - Designed a timestamp-based version-state model to resolve sync conflicts between Notion pages and MongoDB records.
 - Developed a document-linking layer mapping a canonical schedule to multiple nodes for one-to-many synchronization.
- **NTU Pair** Dec 2022 – Feb 2023
 - Built a full-stack matchmaking platform using *vanilla JS*, reaching 1,600+ verified users.
 - Re-architected the system with secure coding practices after mitigating XSS, DoS, and credential-stuffing attacks.

Leadership

- **Founder and President, NTU Learning Optimization Club** Aug 2023 – Jun 2025
 - Researched learning methodologies, delivered 8 nationwide speeches, and advised 200+ students on study habits.
- **Lecturer, NTUEE Student Association Academic Department** Sep 2023 – Dec 2023
 - Organized 25 academic workshops and taught microcontroller programming to electrical engineering undergraduates.
- **Project Lead, NTUEE Virtual Makerspace** May 2023 – Aug 2023
 - Led a 25-member team to build a React/Node.js inventory and reservation system for the university student makerspace.

Skills

- **Languages:** Python, JavaScript/TypeScript, C++, Verilog, MATLAB
- **Frameworks/Tools:** React/Next.js, Node.js, Pandas, PyTorch, MongoDB, Docker, Git
- **Specialties:** System design, algorithm design, full-stack development, formal verification